

Five Python Libraries that Make Data Visualisation Easy

Data visualisation is an inevitable task, considering the prolific growth in the volume and nature of data that applications need to handle. In such a scenario, presenting data in the form of easy-to-comprehend visual representations increases its value. This article provides an introduction to five data visualisation libraries of Python.



The extensibility of Python is one of the main reasons for its popularity among developers from a wide spectrum of domains. Invariably, all of these domains handle large volumes of data, and presenting this data in an easy-to-grasp format has become a mandatory requirement. Python is loaded with options when it comes to visualisation of data. Some of them are listed in Figure 1.

Each of these visualisation libraries has its own salient features. Depending on the context, specific libraries may be adopted for implementation. Moreover, some of these libraries are dependent on other libraries. For example, Seaborn is a statistical data visualisation library that uses Matplotlib, a visualisation library widely used by Python developers. This article presents an overall view of five visualisation libraries of Python, which are: Bokeh, Seaborn, Altair, ggplot and Pygal.

Bokeh

The Bokeh (<http://bokeh.pydata.org/en/latest/>) visualisation library is aimed at building interactive graphical representations. These are rendered in Web browsers. The visualisation rendered using Bokeh is in line with D3.js in terms of style. High performance and interactivity, even with large datasets, are important features of this library.

The official documentation recommends the use of the Anaconda Python distribution to set up Bokeh effortlessly, using the following command:

```
conda install bokeh
```

Nevertheless, Anaconda is not a mandatory requirement to install Bokeh. If the dependencies are satisfied, it can be installed using Pip as well.

As stated earlier, making plots with Bokeh is very simple. The following code snippet is to build a simple line graph and render it in a browser:

```
from bokeh.plotting import figure, output_file, show
```

```
# 1. Load some data in X,Y variables
```

```
x = [1, 2, 3, 4, 5]
```

```
y = [23, 15, 7, 12, 21]
```

```
# 2. Specify the name of the output HTML file
```

```
output_file("lines.html")
```

```
#3. create a new plot with a title and axis labels
```

```
p = figure(title="Bokeh Demo for OSFY", x_axis_label='x', y_
```